

Jasveer Dhillon

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EXPERIENCE

York University – NSERC USRA

May 2025 – Aug. 2025

Undergraduate Researcher (Supervisor: Prof. Simone Pisana)

Toronto, ON

- Conducted research on *Using Graphene as a Material for Tritium and Deuterium Separation in Water* under the NSERC Undergraduate Student Research Award
- Fabricated graphene oxide (GO) filters and prepared samples for structural and material characterization
- Trained to operate a Raman spectroscope; collected and analyzed spectral data to estimate degree of oxidation in GO membranes
- Processed raw spectroscopy data using Python and used SciPy for curve fitting and peak analysis to generate plots
- Presented research findings at a summer research conference

PROJECTS

Fishy – BASIC to C Translator | Python | GitHub

- Developed a source-to-source compiler translating a subset of BASIC into ANSI C
- Implemented tokenizer, recursive-descent parser, and structured C code emitter
- Implemented support for variables, arithmetic expressions, control flow (IF, WHILE), I/O operations, labels, and GOTO statements
- Strengthened understanding of compiler design concepts including tokenization, syntax trees, and source-to-source translation

Autonomous SumoBot | Arduino, Embedded C/C++

- Designed and built a fully autonomous sumo robot from scratch, including chassis design, motor control, and sensor integration
- Developed custom Arduino firmware for real-time opponent detection, edge sensing, and strategic movement control
- Implemented sensor fusion using IR distance sensors and line-detection sensors to prevent ring exit while tracking opponents
- Competed at McMaster University SumoBot Competition and placed **4th out of 40 teams**

EDUCATION

York University

Sept. 2023 – May 2028

B.Sc. Hons. Computer Science

CGPA: 8.06/9

- **Coursework:** Data Structures, Algorithms, Databases, Computer Systems
- **Awards:**
 - * *NSERC Undergraduate Student Research Award (USRA)* – Competitive national research award
 - * *Denise Hobbins Prize* – Highest average in first-year physics courses
 - * *Deans Honour Roll Member*

SKILLS

Programming: C/C++, Python, Java, Bash, x86_64/ARM Assembly, SQL (MySQL)

Frontend: HTML/CSS, JavaScript, jQuery, React.js

Tools: Visual Studio, JetBrains IDEs, Jupyter Notebooks, VIM, GCC/Clang, Linux

Libraries: NumPy, Pandas, Matplotlib, SciPy